

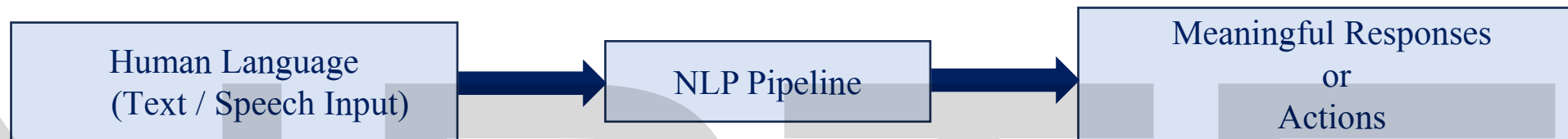
Foundations of Natural Language Processing

Contents

- Introduction to NLP
- Applications of NLP
- NLP Pipeline
- Text Preprocessing
 - Tokenization
 - Stop Word Removal
 - Lemmatization
- Text Representation
 - Word2Vec Word Embeddings
 - GloVe Embedding
- Challenges with NLP

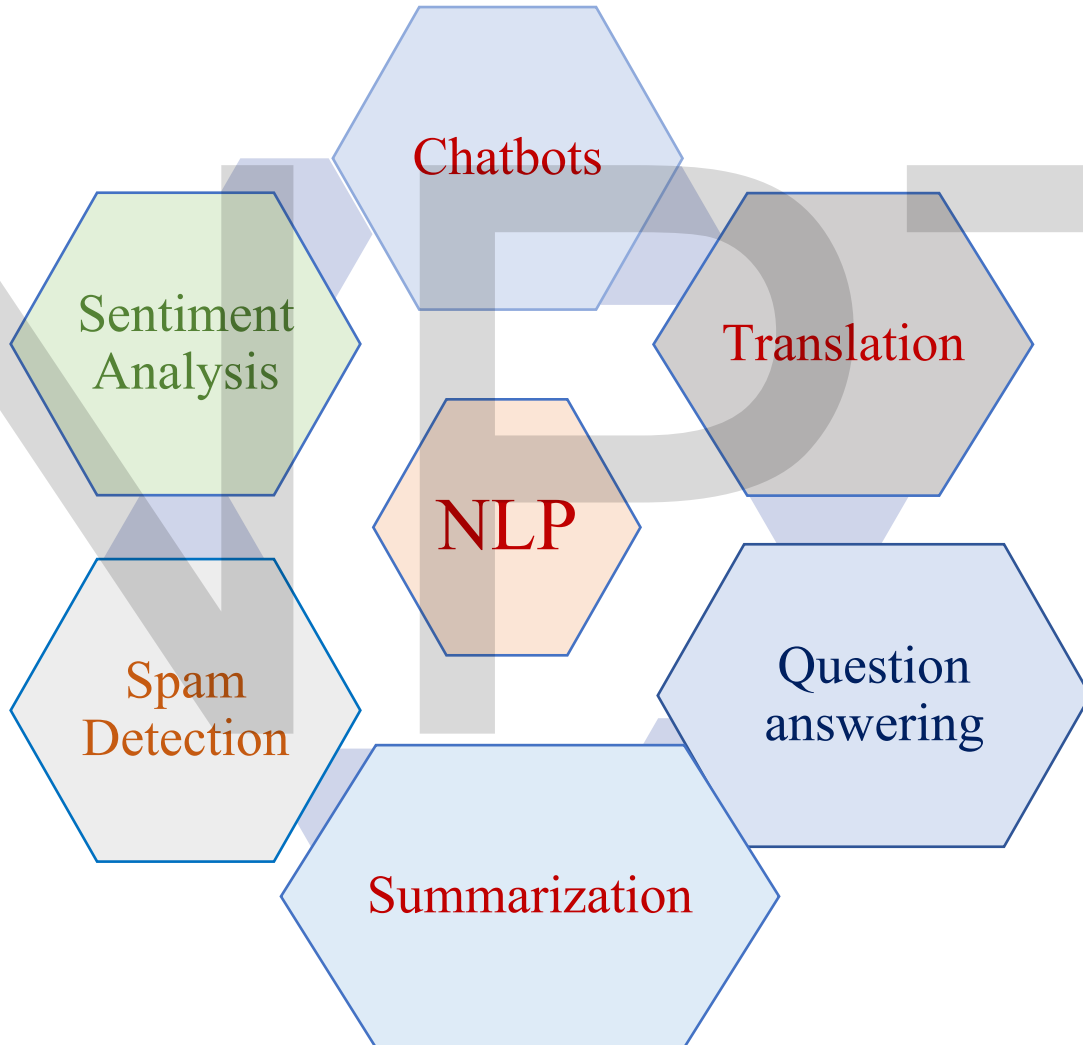
Natural Language Processing (NLP)

Natural language processing (NLP) is a technology that allows computers to interpret, manipulate, and comprehend human language.



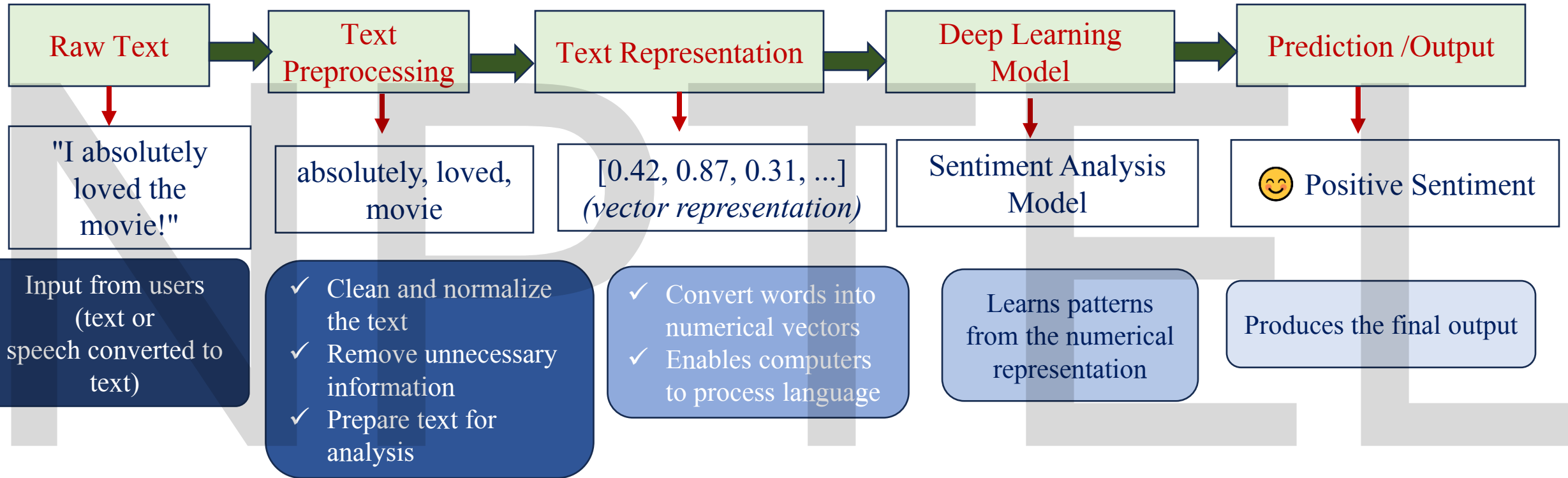
Input	Output	Action
"Is this email spam?"	Yes	Spam Detection
"The movie was amazing!"	Positive Sentiment	Sentiment Analysis
"Hello"	"Hola"	Machine Translation
"What is Generative AI?"	Generative AI is a branch of AI that learns patterns from existing data and generates new content that is similar, but not identical, to the training data.	Question Answering

Applications of NLP



Whenever a computer processes, understands, or generates human language, NLP is at work.

NLP Pipeline



Text Preprocessing

Process of cleaning and transforming raw text into a structured format that can be easily analyzed by machine learning and deep learning models.



Raw Text

"The Quick Brown Foxes are Running!!! Visit <https://example.com> 😊"

After Preprocessing

quick brown fox run

One of the most fundamental preprocessing steps is **tokenization**, where we break text into smaller units called tokens

Tokenization

- Tokenization is the process of splitting text into smaller units called *tokens*.
- A token can be a word, a sentence, a subword, or even a character, depending on the NLP application.

Word Tokenization

"I love learning Natural Language Processing."

'I'	'love'	'learning'	'Natural'	'Language'	'Processing'
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Sentence Tokenization

"NLP is fascinating. It enables computers to understand language. It powers ChatGPT."

'NLP is fascinating'

'It enables computers to
understand language'

'It powers ChatGPT'

Tokenization

Subword Tokenization

"unbelievable"

‘un’ ‘believ’ ‘able’

(used in modern LLMs such as BERT and GPT)

Character Tokenization

"NLP"

‘N’ ‘L’ ‘P’

- ✓ Tokenization converts raw text into smaller meaningful units (tokens),
- ✓ Making it possible for computers to process human language effectively.

Stop Word Removal

- Stop words are frequently occurring words like "the", "is", "are", "of", "and"

Original Sentence

The students are learning Natural Language Processing.

After Stop Word Removal

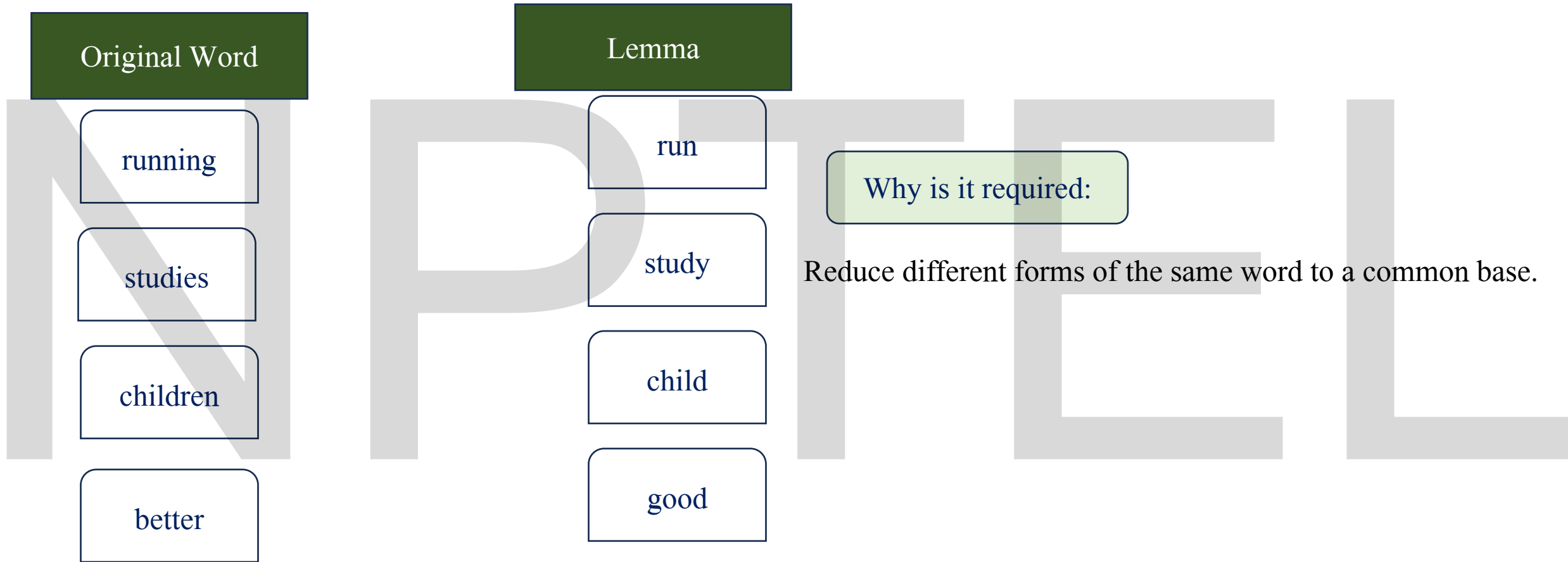
students learning Natural Language Processing

Why is it required:

Stop words usually carry little semantic meaning and can often be removed to reduce noise in the text.

Lemmatization

- Process of converting a word to its base or dictionary form (called the *lemma*) while preserving its meaning.



After preprocessing, how does a computer actually understand words?

Text Representation

Core Idea

- ✓ Machines do not understand words or sentences.
- ✓ Text must be converted into numbers before it can be used by machine learning or deep learning models.

Traditional Techniques

- ✓ One-Hot Encoding
- ✓ Bag of Words (BoW)
- ✓ TF-IDF

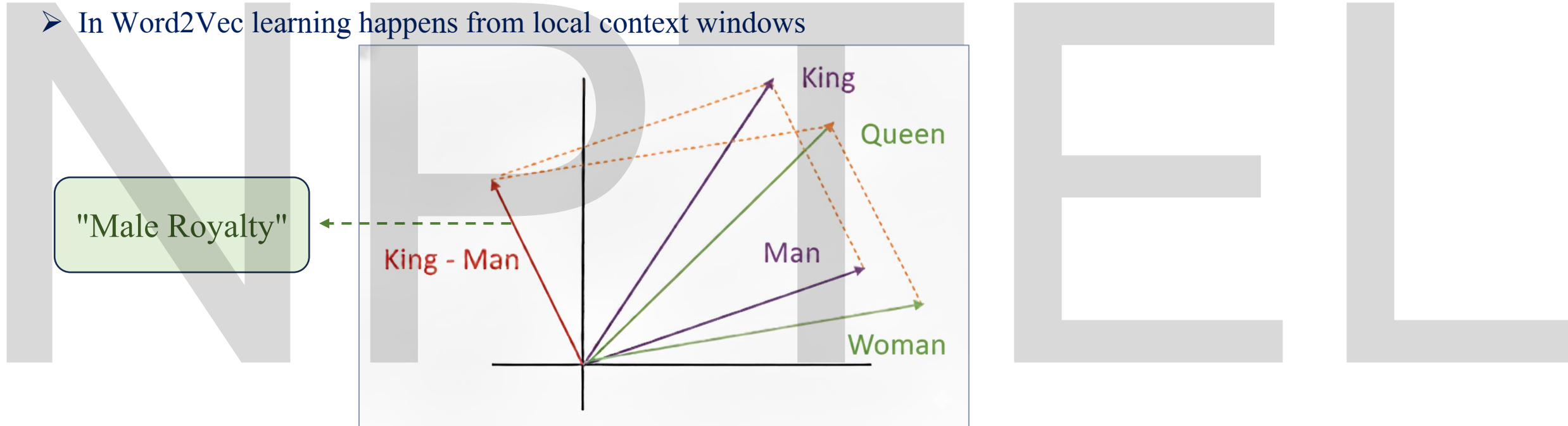
Modern Deep Learning Techniques

- ✓ Word Embeddings (Word2Vec, GloVe)
- ✓ Contextual Embeddings (BERT, GPT)

Word2Vec Word Embeddings

Core Idea

- Words that appear in similar contexts are represented by similar vectors in the embedding space.
- In Word2Vec learning happens from local context windows



The **direction** and **length** of the vector encode the semantic properties of the word.

$$(\text{King} - \text{Man}) + \text{Woman} \approx \text{Queen}$$

GloVe Embedding

Core idea

- ✓ Words that frequently occur together in a large corpus are likely to have similar meanings.
- ✓ Learn Dense Word Embeddings using Word Co-occurrence Matrix

Co-occurrence Matrix

Word	King	Queen	Man	Woman
King	0	120	250	80
Queen	120	0	75	240
Man	250	75	0	110
Woman	80	240	110	0

Higher co-occurrence values indicate stronger semantic relationships between words.

Challenges in NLP

Discrete Nature of
Text data

Sky and stars
Red and pink



Some relation is there among these pair of words
but words doesn't tell anything about the similarity

Evolving nature of
Natural Language

New words gets added to the dictionary from time to time(ex: rizz)

Ambiguity/Vagueness

Never tasted a pizza like this before

Capturing long range
dependency

Animal did not cross the road as it was (wide/tired)

NPTEL

Summary

NPTTEL

Next Session

Sequential Modeling with RNNs and LSTMs